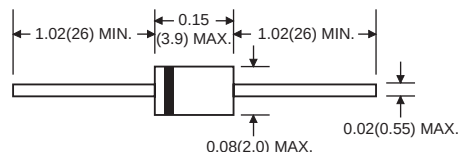




Features

- ✧ Fast switching speed
- ✧ General purpose rectification
- ✧ Silicon epitaxial planar construction



Mechanical Data

- ✧ Case: DO-35
- ✧ Leads: Pure tin plated lead free, Solderable per MIL-STD-202, Method 208
- ✧ Polarity: Cathode band
- ✧ Marking: Type number
- ✧ Weight: 0.13 grams (approx.)

Dimensions in inches and (millimeters)

Maximum Ratings and Electrical Characteristics

Rating at 25 °C ambient temperature unless otherwise specified.

Maximum Ratings

Type Number	Symbol	1N4148	Units
Repetitive Peak Reverse Voltage	V _{RRM}	100	V
Reverse Voltage	V _R	75	V
Peak Forward Surge Current tp=1us	I _{FSM}	2	A
Repetitive Peak Forward Current	I _{FRM}	500	mA
Forward Current	I _F	300	mA
Average Forward Current V _R =0	I _{FAV}	150	mA
Power Dissipation I=4mm, T _L =45 °C I=4mm, T _L ≤25 °C	P _V	440	mW
	P _V	500	mW
Junction Ambient I=4mm, T _L = constant	R _{θJA}	350	K/W
Operating and Storage Temperature Range	T _J , T _{STG}	-65 to + 200	°C

Electrical Characteristics

Type Number	Symbol	Min	Max	Units
Forward Voltage @ IF=10mA	V _F	-	1.0	V
Peak Reverse Current VR=75V VR=20V, T _J =150 °C VR=20V	I _R	-	5	uA
			50	uA
			25	nA
Breakdown Voltage I _R =100uA, tp/T=0.01, tp=0.3ms	V _(BR)	100	-	V
Capacitance VR=0, f=1.0MHz, V _{HF} =50mV	C _j	-	4.0	pF
Rectification Efficiency VHF=2V, f=100MHz	T _{lr}	45	-	%
Reverse Recovery Time IF=I _R =10mA, I _R =1mA IF=10mA, VR=6V, I _R =0.1x I _R , R _L =100Ω	trr	-	8.0	nS
	trr	-	4.0	nS

RATINGS AND CHARACTERISTIC CURVES (1N4148)

FIG.1- ADMISSIBLE POWER DISSIPATION VS AMBIENT TEMPERATURE

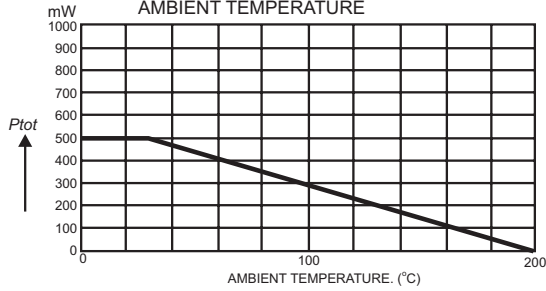


FIG.2- TYPICAL REVERSE CHARACTERISTICS

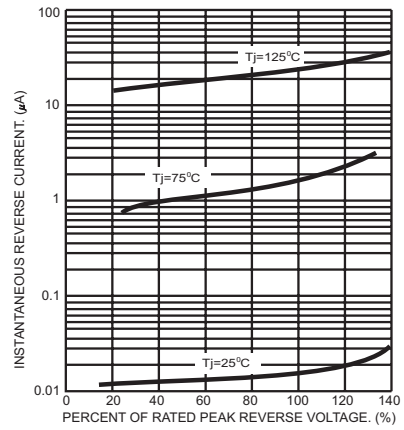


FIG.3- MAXIMUM NON-REPETITIVE FORWARD SURGE CURRENT

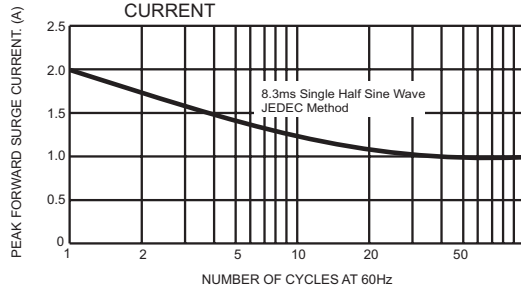


FIG.5- FORWARD CHARACTERISTICS CURVE

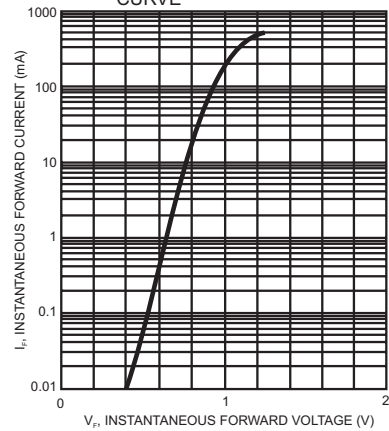


FIG.4- TYPICAL JUNCTION CAPACITANCE

